Introduction to Estimation Theory

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Key Concepts

1) Estimation theory concerns the problem of using the observed data to infer the value of an unknown parameter describing (partially) the data.

    a) How do we find a function of the data that estimates the unknown parameter?

    b) How do we characterize the quality of our estimate?

2) Metrics of estimator quality:

a) Bias is the difference between the mean of the estimate and the true value of the parameter. An unbiased estimator has zero bias.

b) Variance is the expected squared difference between the estimate and its average value.

c) Mean-squared error (MSE) is the expected squared difference between the estimate and the true value of the parameter. MSE is the sum of the variance and the square of the bias.

Video Screenshots

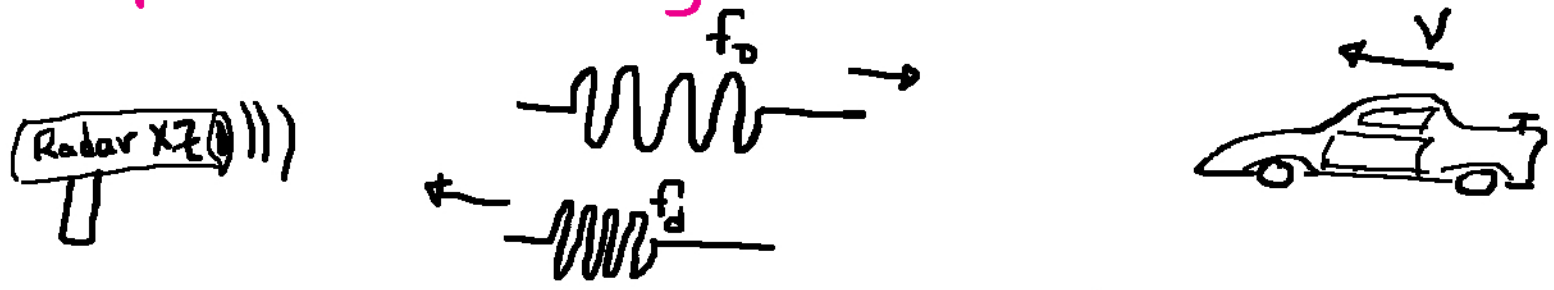
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Parameter estimation - probability distribution of data $\underline{x}$ depends on an unknown parameter α

$$\underline{x} \sim f(\underline{x}; \alpha)$$

Estimate α from observed $\underline{x}$

Example: Velocity estimation



Known: f_0 Estimate f_d to find v

$$x[n] = A\cos(2\pi f_d n + \phi) + w[n]$$

noise

$$\hat{f}_d = g(x[0], x[1], \ldots x[N-1])$$

Let $\hat{\underline{\alpha}} = g(\underline{x})$ be the estimate of $\underline{\alpha}$

1) How do we find $g(\cdot)$?
2) How "good" is $g(\cdot)$?

$\underline{x}$ is random, so $\hat{\underline{\alpha}}$ is random!

How concentrated is the pdf of $\hat{\underline{\alpha}}$ about the true $\underline{\alpha}$?

In general, let $\underline{\alpha}$ be a vector (multiple parameters)

1) Bias

$$b(\hat{\underline{\alpha}}) = E\{\hat{\underline{\alpha}}\} - \underline{\alpha}$$

$E\{\hat{\underline{\alpha}}\}$ ← mean of estimate

If $b(\hat{\underline{\alpha}}) = \underline{0}$, we say $\hat{\underline{\alpha}}$ is unbiased

2) (Co)variance

$$\underline{C}(\hat{\underline{\alpha}}) = E\{(\hat{\underline{\alpha}} - E\{\hat{\underline{\alpha}}\})(\hat{\underline{\alpha}} - E\{\hat{\underline{\alpha}}\})^T\}$$

$$\text{Variance}\{[\hat{\underline{\alpha}}]_i\} = [\underline{C}(\hat{\underline{\alpha}})]_{ii}$$



3) Mean-squared Error

$$MSE\{[\hat{\underline{\alpha}}]_i\} = E\{[\hat{\underline{\alpha}} - \underline{\alpha}]_i^2\}$$

$$= \text{Variance}\{[\hat{\underline{\alpha}}]_i\} + (\text{bias}\{[\hat{\underline{\alpha}}]_i\})^2$$

In general

$$MSE\{\hat{\underline{\alpha}}\} = \underline{C}(\hat{\underline{\alpha}}) + \underline{b}(\hat{\underline{\alpha}})\,\underline{b}^T(\hat{\underline{\alpha}})$$

Example: given $x_1, x_2, x_3 \sim N(m, \sigma^2)$ estimate m, σ^2

8:15

$$\hat{m} = \frac{1}{3}(x_1 + x_2 + x_3)$$

$$\hat{\sigma}^2_{bias} = \frac{1}{3}\sum_{i=1}^{3}(x_i - \hat{m})^2$$

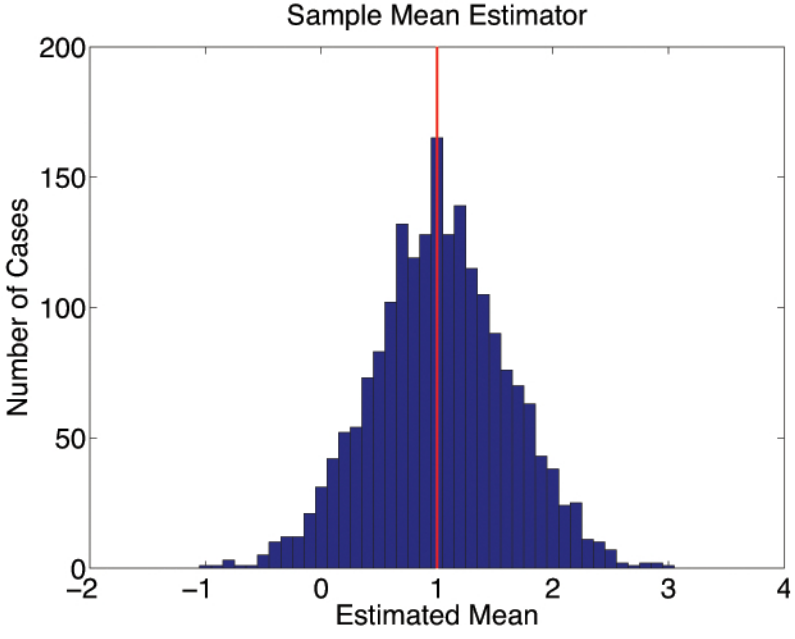
$$\hat{\sigma}^2_{unbias} = \frac{1}{2}\sum_{i=1}^{3}(x_i - \hat{m})^2$$

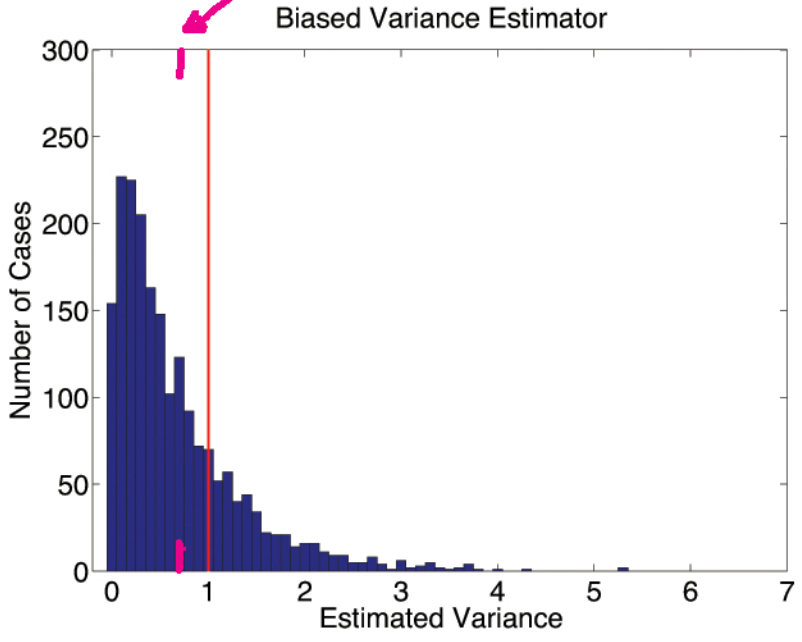
$m = 1$
$\sigma^2 = 1$: 2000 times

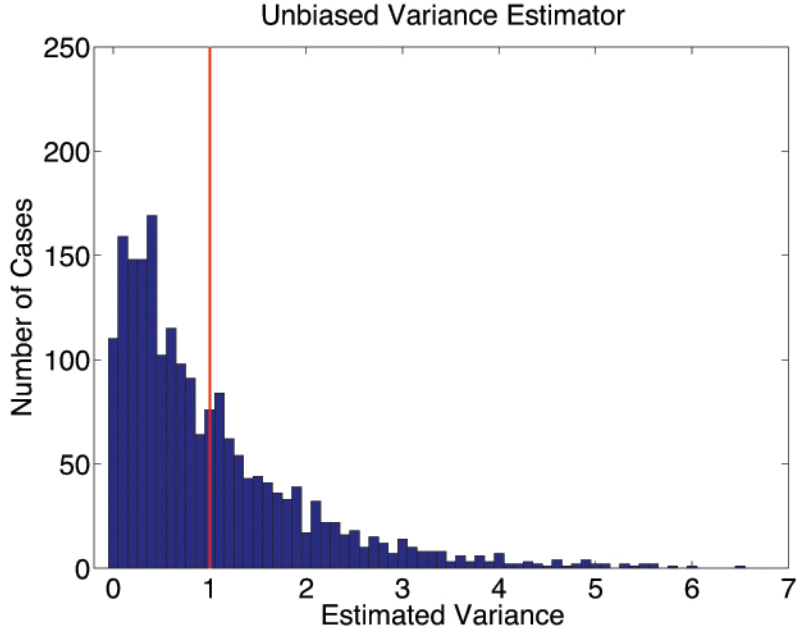
$E\{\hat{m}\} \approx 1.02$

$E\{\hat{\sigma}^2_{bias}\} \approx 0.67$

$E\{\hat{\sigma}^2_{unbias}\} \approx 1.01$







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